COMP 3605 Tutorial 3

Association Analysis Basics

Introduction to Data Analytics, UWI St. Augustine

Semester I, 2026/2027

How to use this sheet. Attempt every question before your tutorial session, showing all working. Solutions with step-by-step explanations will be released afterwards.

Formulas you will need

$support(A \Rightarrow B) = P(A \cup B)$ (fraction of transactions containing both); $confidence(A \Rightarrow B) = P(B \mid A) = \dfrac{support_count(A \cup B)}{support_count(A)}$.

A rule is **strong** if support $\geq$ min_sup and confidence $\geq$ min_conf. Apriori property: every subset of a frequent itemset is frequent.

$lift(A, B) = \dfrac{P(A \cup B)}{P(A)P(B)}$ (> 1 positive, < 1 negative correlation, $= 1$ independent); $all_conf = \dfrac{sup(A \cup B)}{\max\{sup(A), sup(B)\}}$; $max_conf = \max\{P(A|B), P(B|A)\}$; $Kulc = \frac{1}{2}(P(A|B) + P(B|A))$; $cosine = \dfrac{P(A \cup B)}{\sqrt{P(A)P(B)}}$.

Question 1: Warm-up

(a) In one sentence each, define: itemset, frequent itemset, association rule.

(b) Why does mining proceed in two steps (find frequent itemsets first, generate rules second), and which step is the expensive one?

(c) State the Apriori property and explain how Apriori uses it to prune candidates. Include a concrete example: if $\{beer, nuts\}$ is infrequent, what do we know about $\{beer, nuts, diapers\}$?

Question 2: Apriori on one data set

A database has five transactions. Let min_sup = 60% and min_conf = 80%.

TID	items
T100	{M, O, N, K, E, Y}
T200	{D, O, N, K, E, Y}
T300	{M, A, K, E}
T400	{M, U, C, K, Y}
T500	{C, O, O, K, I, E}

(a) Find all frequent itemsets using **Apriori**. Show every candidate set C_k and frequent set L_k.

(b) List all **strong association rules** (support $\geq$ 60%, confidence $\geq$ 80%) that can be generated from the frequent 2-itemsets and 3-itemsets. For each rule give its support and confidence.

Question 3: Apriori at two thresholds

Five transactions:

TID	items
1	{Bread, Butter, Milk}
2	{Eggs, Milk, Yogurt}
3	{Bread, Cheese, Eggs, Milk}
4	{Eggs, Milk, Yogurt}
5	{Cheese, Milk, Yogurt}

Using Apriori, find all frequent itemsets for

(a) min_sup = 0.3;

(b) min_sup = 0.5.

In each case, first convert min_sup to a minimum support *count*.

Question 4: Vertical data format

Using the same database as Question 3:

(a) Rewrite the database in vertical format: for each item, its TID-set.

(b) Find all frequent itemsets for min_sup = 0.4 by intersecting TID-sets. Show the intersections.

(c) Repeat for min_sup = 0.6.

(d) What is the advantage of the vertical representation when counting the support of k-itemsets?

Question 5: Prove it

Let Z be an itemset in database D.

(a) Prove that every nonempty subset of a frequent itemset is frequent.

(b) Prove that for any nonempty subset X of Z, $support(X) \geq support(Z)$.

(c) Let $X' \subset X \subseteq Z$, both nonempty. Prove that $confidence(X' \Rightarrow (Z - X')) \leq confidence(X \Rightarrow (Z - X))$.

Question 6: Strong but negatively correlated

150 customers; 100 bought item A, 105 bought item B, 65 bought both.

	B	$\bar{B}$	total
A	65	35	100
$\bar{A}$	40	10	50
total	105	45	150

(a) With min_sup = 40% and min_conf = 60%, is $A \Rightarrow B$ a strong rule?

(b) Compute the lift of A and B. Are they independent, positively correlated, or negatively correlated?

(c) In one or two sentences: what lesson does this example teach about the support-confidence framework?

Question 7: Hot dogs and hamburgers

5000 transactions:

	hotdogs	$\overline{hotdogs}$	total
hamburgers	2000	500	2500
$\overline{hamburgers}$	1000	1500	2500
total	3000	2000	5000

(a) With min_sup = 25% and min_conf = 50%, is $hotdogs \Rightarrow hamburgers$ a strong rule?

(b) Is the purchase of hot dogs independent of the purchase of hamburgers? If not, what kind of correlation exists?

(c) Compute all_conf, max_conf, $Kulc$, and $cosine$ for hot dogs and hamburgers.

Question 8: Closed and maximal itemsets

Four transactions: $T_1 = \{a, b, c\}$, $T_2 = \{a, b, c\}$, $T_3 = \{a, b\}$, $T_4 = \{a\}$. Min_sup count $= 2$.

(a) List every frequent itemset with its support count.

(b) An itemset is *closed* if no proper superset has the same support. Which frequent itemsets are closed?

(c) An itemset is *maximal* if no proper superset is frequent. Which are maximal?

(d) Why are closed frequent itemsets a *lossless* compression of the frequent itemsets, while maximal ones are lossy?

End of Tutorial 3. Attempt everything before looking at the solutions.